Data Visualization

Introduction to data visualization, Visualization foundations

KESIC DCA

Introduction to data visualization

- Data visualization is a powerful tool for enhancing understanding and communication of complex data. It involves representing data in a graphical or pictorial form, making it easier to understand and interpret.
- The importance of data visualization in effectively communicating and analyzing data to provide insights into the various types of data visualization tools and techniques available.

Overview of Data Visualization Tools :

- Data visualization tools can be broadly classified into three categories: spreadsheets, data visualization software, and programming libraries.

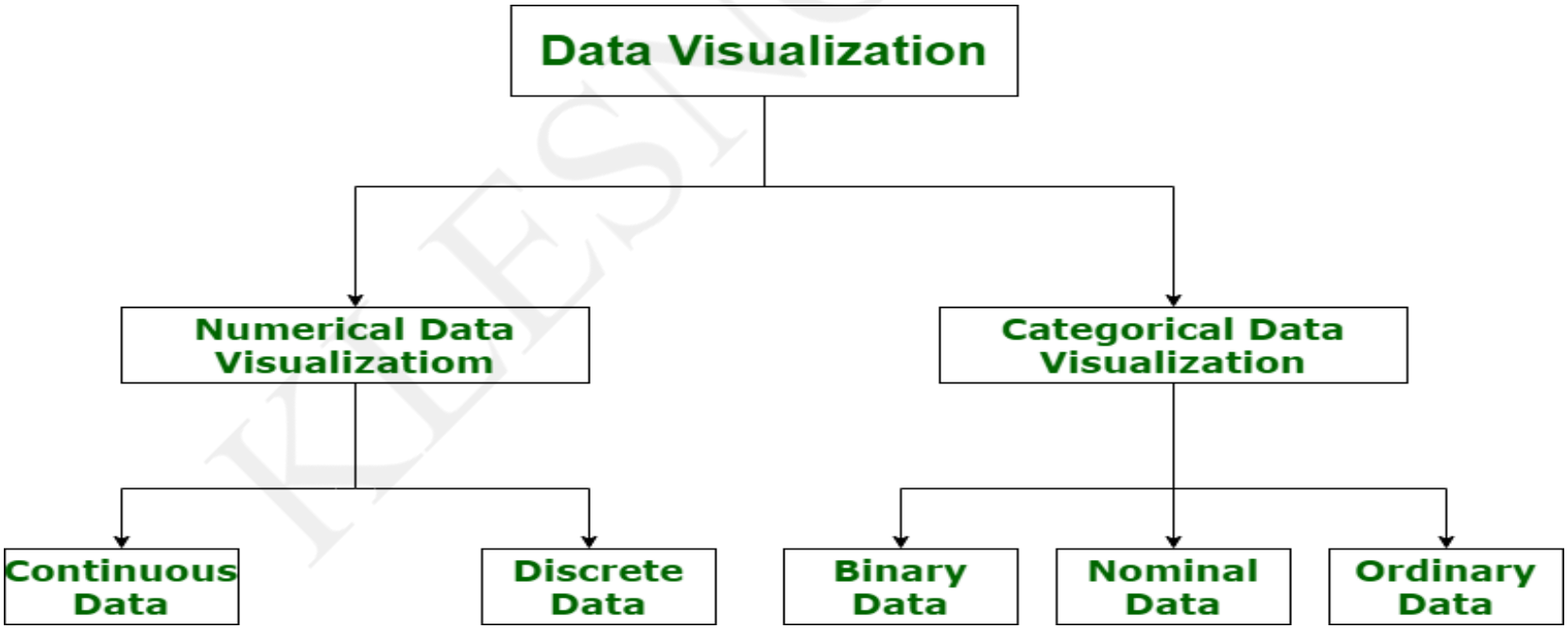
KTESNC BCA

- **Spreadsheets** - Spreadsheets, such as Microsoft Excel and Google Sheets, are one of the most common data visualization tools used in various domains. They provide basic data visualization capabilities, such as bar charts, line graphs, and scatter plots.
- **Data Visualization Software** - Data visualization software is a specialized tool designed for data visualization and analysis. Examples of data visualization software include Tableau, QlikView, and Power BI. These tools provide advanced data visualization capabilities, including interactive dashboards, heat maps, and network diagrams.
- **Programming Libraries** - Programming libraries, such as Matplotlib, ggplot2, and D3.js, are a type of data visualization tool that can be used to create custom data visualizations.

KLEBCA

CATEGORIES OF DATA VISUALIZATION

- Data visualization is very critical to market research where both numerical and categorical data can be visualized that helps in an increase in impacts of insights and also helps in reducing risk of analysis paralysis. So, data visualization is categorized into following categories :



KLESIC BCA

Numerical Data :

- Numerical data is also known as Quantitative data. Numerical data is any data where data generally represents amount such as height, weight, age of a person, etc.
- Numerical data is categorized into two categories :

Continuous Data –

- It can be narrowed or categorized (Example: Height measurements).

Discrete Data –

- This type of data is not "continuous" (Example: Number of cars or children's a household has).

Examples are Pie Charts, Bar Charts, Averages, Scorecards, etc.

KLESNC BCA

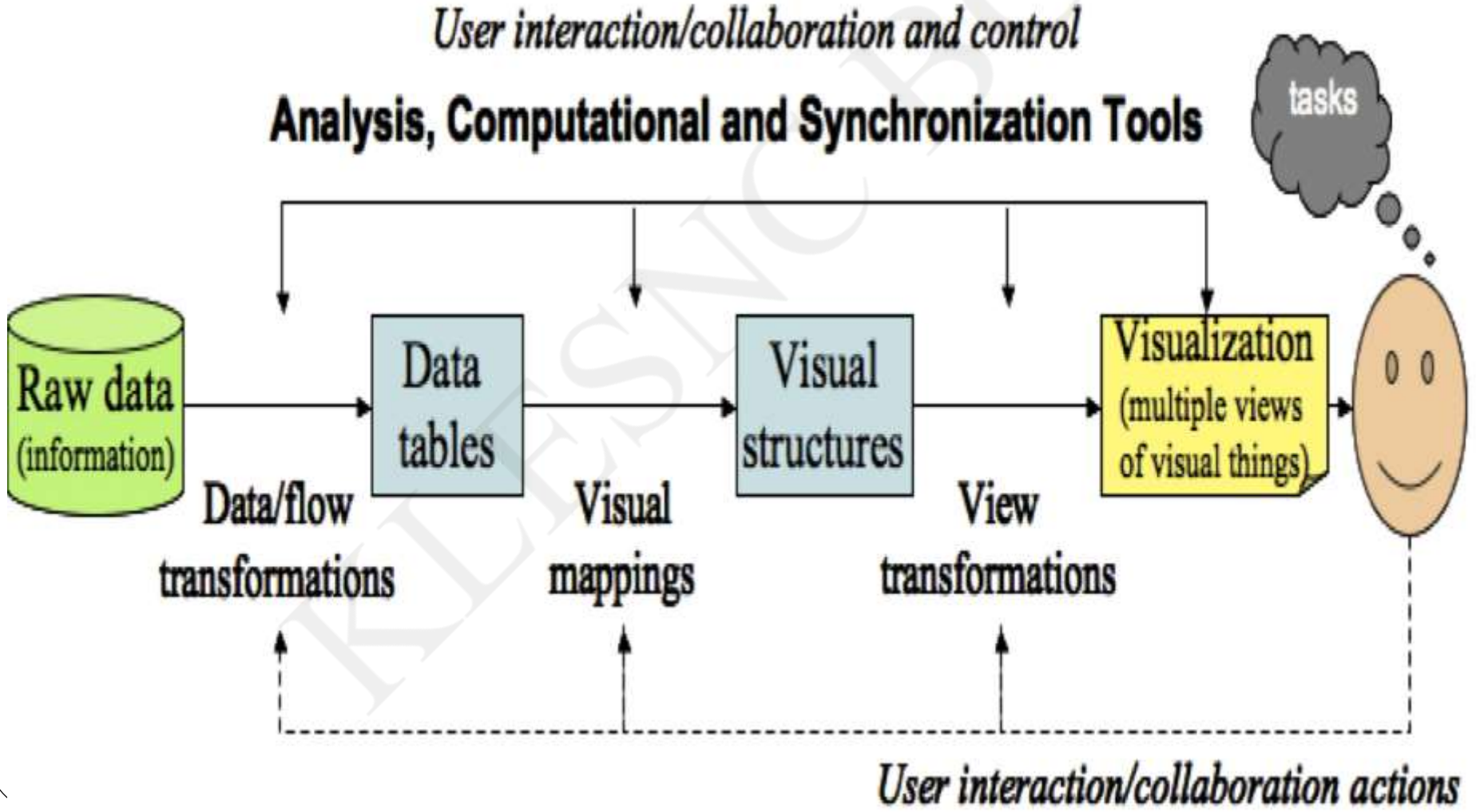
Categorical Data :

- Categorical data is also known as Qualitative data. Categorical data is any data where data generally represents groups. It simply consists of categorical variables that are used to represent characteristics such as a person's ranking, a person's gender, etc.
  - **Binary Data –**
    - In this, classification is based on positioning (Example: Agrees or Disagrees).
  - **Nominal Data –**
    - In this, classification is based on attributes (Example: Male or Female).
  - **Ordinal Data –**
    - In this, classification is based on ordering of information (Example: Timeline or processes).

KLESNC BCA

Foundations of Data Visualization

- **Visualization Process Details:**



- **Data preprocessing and transformation:** The starting point is to process the raw data into something usable by the visualization system. The first part is to make sure that the data are mapped to fundamental data types for computer ingestion.
- The second step entails dealing with specific application data issues such as missing values, errors in input, and data too large for processing. The data may be simulated or sampled.
- **Mapping for visualizations :** Once the data are clean, we can decide on a specific visual representation. This requires representation mappings:

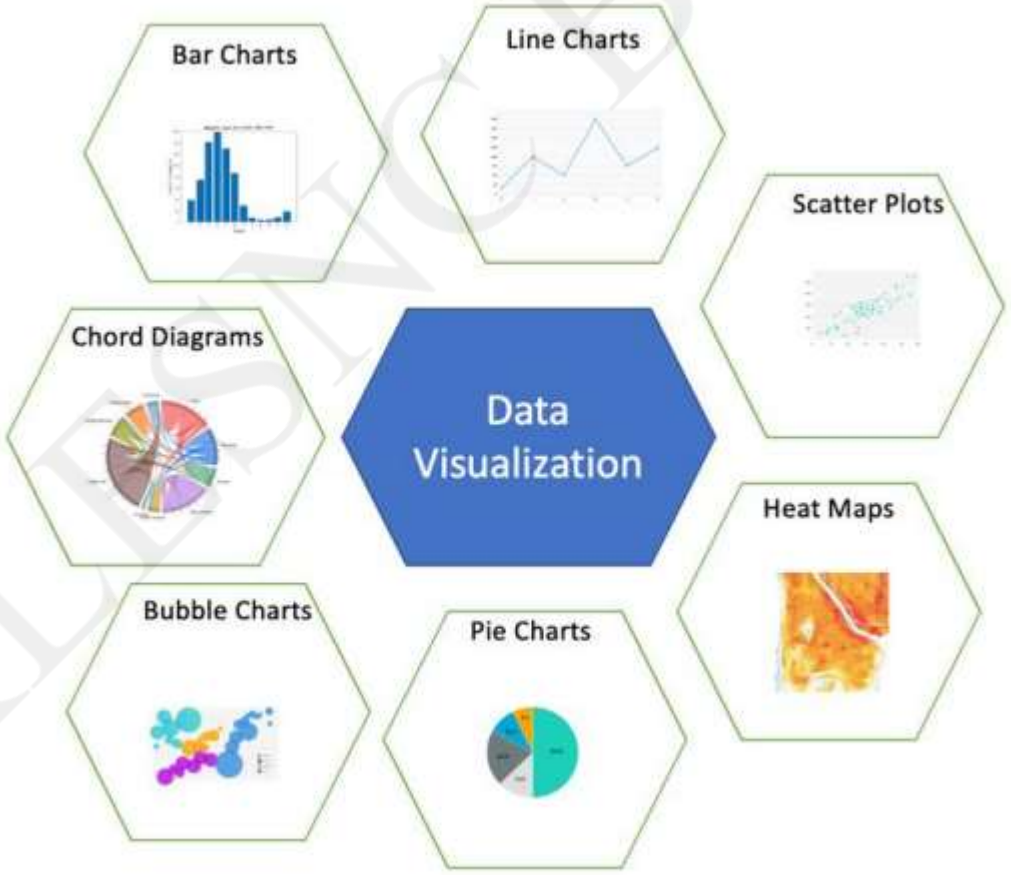
KLESNCBCA

- **Rendering transformations :**The final stage involves mapping from geometry data to the image. This includes interfacing with a computer graphics Application Programmer's Interface (API). We need to select the viewing parameters, shading technique if 3D, device transformations (for display, printers, . . .). This stage of the pipeline is very dependent on the underlying graphics library.
- **Expressiveness :** An expressive visualization presents all the information, and only the information Expressiveness thus measures the concentration of information. Given information that we actually display to the user.
- **Effectiveness :**A visualization is effective when it can be interpreted accurately and quickly and when it can be rendered in a cost-effective manner. Effectiveness thus measures a specific cost of information perception.

KUESNC BCA

Data visualization techniques

- Data visualization techniques are used to represent data in a graphical or pictorial form, making it easier to understand and interpret. There are several types of data visualization techniques, including:



KLESNCBCA

- Bar charts are useful for comparing data across categories, where the length of each rectangular bar represents the magnitude of a particular data point.
- Line graphs are used to visualize trends over time, with a series of points connected by lines where each point represents a data point for a particular period.
- Scatter plots, on the other hand, are used to visualize the relationship between two variables. A series of points represents a pair of values for two variables.
- Heat maps are helpful in visualizing the distribution of values in a matrix or table. The colour of each cell in a coloured grid represents the magnitude of a particular data point.
- Network diagrams are useful in visualizing relationships between entities, consisting of nodes representing entities and edges representing relationships, where the size and colour of nodes and edges can represent data attributes.

KLESNC BCA

Why Data visualization is important

- The importance of data visualization is simple: it helps people see, interact with, and better understand data. Whether simple or complex, the right visualization can bring everyone on the same page, regardless of their level of expertise.

1. Analyzing the Data in a Better Way

- Analyzing reports helps business stakeholders focus on the areas that require attention. The visual mediums help analysts understand the key points needed for their business.

2. Faster Decision Making

- If the data communicates well, decision-makers can quickly take action based on the new data insights, accelerating decision-making, and business growth simultaneously.

3. Making Sense of Complicated Data

- Data visualization allows business users to gain insight into their vast amounts of data. It benefits them to recognize new patterns and errors in the data. Making sense of these patterns helps the users pay attention to areas that indicate red flags or progress.

Different types of visualizations

General Types of Visualizations:

- **Chart:** Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map.
- **Table:** A set of figures displayed in rows and columns.
- **Graph:** A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other, usually along two axes at a right angle.
- **Geospatial:** A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations.
- **Infographic:** A combination of visuals and words that represent data. Usually uses charts or diagrams.
- **Dashboards:** A collection of visualizations and data displayed in one place to help with analyzing and presenting data.

KLE SNC BCA

The Benefits of Data Visualization :

1. Discovers trends in data:

- Data visualization allows us to identify patterns, correlations, and trends in large datasets more effectively than poring over raw numbers.

2. Puts the data into the correct context:

- Solves data inefficiencies and absorbs vast amounts of data presented in visual formats. Visualization helps in understanding the significance and context of data points by presenting them in a meaningful way.
- Identifies errors and inaccuracies in data quickly.

3. Provides a perspective on data:

- Increases the speed of decision-making.
- Access real-time information and assist in management functions.
- Explore business insights and achieve business goals in the right direction
- Optimize and instantly retrieve data via tailor-made reports.
- Stay on top of the game by discovering the latest trends.

KLESNC BCA

4. **Data visualization saves time**

- Visualizations allow for a quicker understanding of complex datasets, reducing the time needed to comprehend and analyze information.

5. Tells a data story:

- Visualization helps in crafting a narrative around the data, making it easier for others to understand and engage with the information. It promotes storytelling and conveys the right message to the audience.

DISADVANTAGES OF DATA VISUALIZATION:

- when viewing a visualization with many different data points, it's easy to make an inaccurate assumption. Or sometimes the visualization is just designed wrong so that it's biased or confusing.
- Biased or inaccurate information.
- Correlation doesn't always mean causation.
- Core messages can get lost in translation.

KLESNCBCA

POWER BI

What is Power BI?

- Power BI is one of the most popular Data Visualization and Business Intelligence tool. The Power BI tool is the collection of apps, data connectors, and software services which are used to get the data from different data sources, transforms data, and produces useful reports.
- Power BI services are based on SaaS and mobile Power BI apps that are available for different platforms. These set of services are used by the business users to consume data and to build Power BI reports.

KLESNCBCA

Reasons to Use Power BI:

- It allows real-time dashboard updates.
- It provides secure and reliable connections to the data sources in the cloud.
- It allows data exploration using a natural language query.
- Power BI provides a hybrid configuration, quick deployment, and secure environment.
- It provides features for dashboard visualization regularly updated with the community.
- It provides pre-built dashboards and reports for SaaS solutions.

KLESNC BCA

Power BI – Advantages

1. **Secure Report Publishing:** You can automate setup data refresh and publish reports that allowing all the users to avail the latest information.
2. **No Memory and Speed Constraints:** To Shift an existing BI system into a powerful cloud environment with Power BI embedded eliminates memory. Speed constraints ensure that data is quickly retrievable and analyzed.
3. **No Specialized Technical Support required:** The Power BI provides quick inquiry and analysis without the need for specialized technical support. It also supports a powerful natural language interface and the use of intuitive graphical designer tools.
4. **Simple to Use:** Power BI is simple to use. Users can easily find it only on behalf of a short learning curve.

KJESNCBCA

5. **Constant innovation:** The Power BI product is updated in every month with new functions and features.
6. **Rich, personalized dashboard:** The crowning feature of Power BI is the information dashboards that can be customized to meet the exact need of any enterprise. You can easily embed the dashboards, and BI reports in the applications to provide a unified user experience.

KLESNC BCA

Power BI Disadvantages

1. Dashboard and reports are only shared with the users who are having the same email domains.
2. Power BI will not merge imported data that is accessed from real-time connections.
3. Power BI only accepts the file size maximum 250Mb and the zip file which is compressed by the data of the x-velocity in-memory database.
4. Dashboard never accepts or pass user, account, or any other entity parameters.
5. Very few data sources permit real-time connections to power BI reports and dashboards.

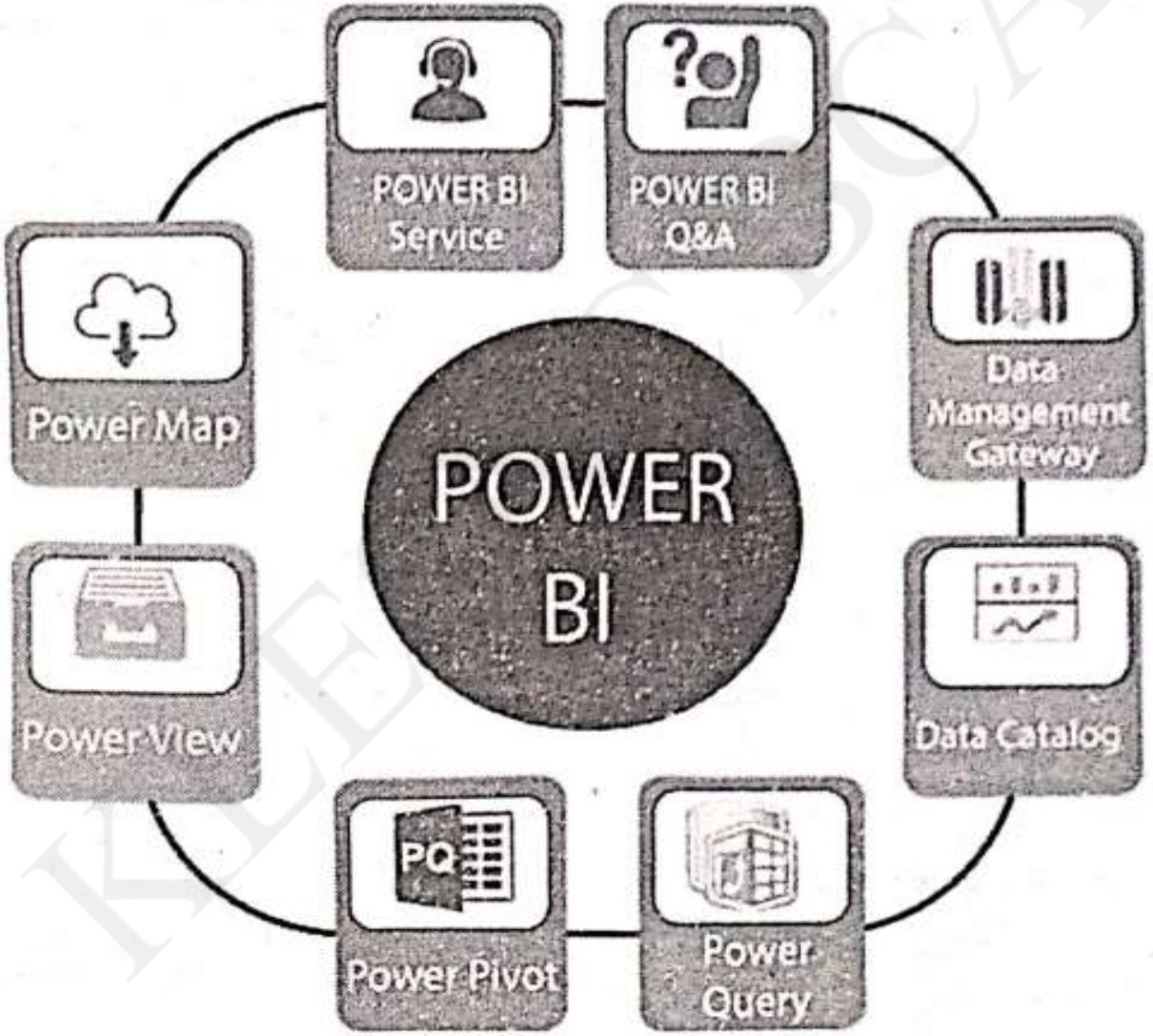
KLESNC BCA

Scalable Options

- Power BI, a business analytics service by Microsoft, offers several scalable options to meet the needs of organizations of various sizes.
- **Power BI Desktop:** This is a free, standalone application for designing and creating Power BI reports and dashboards on a local machine. It's suitable for individual users and small-scale projects.
- **Power BI Service:** This is a cloud-based service that allows you to publish, share, and collaborate on Power BI reports and dashboards.
- **Power BI Pro:** This is a subscription-based license that allows individual users to create and share reports and dashboards. It is suitable for small to medium-sized businesses.
- **Power BI Premium:** This is designed for larger enterprises with more extensive data needs. It offers dedicated cloud capacity and enhanced performance.

KLESIC BCA

Power BI Components

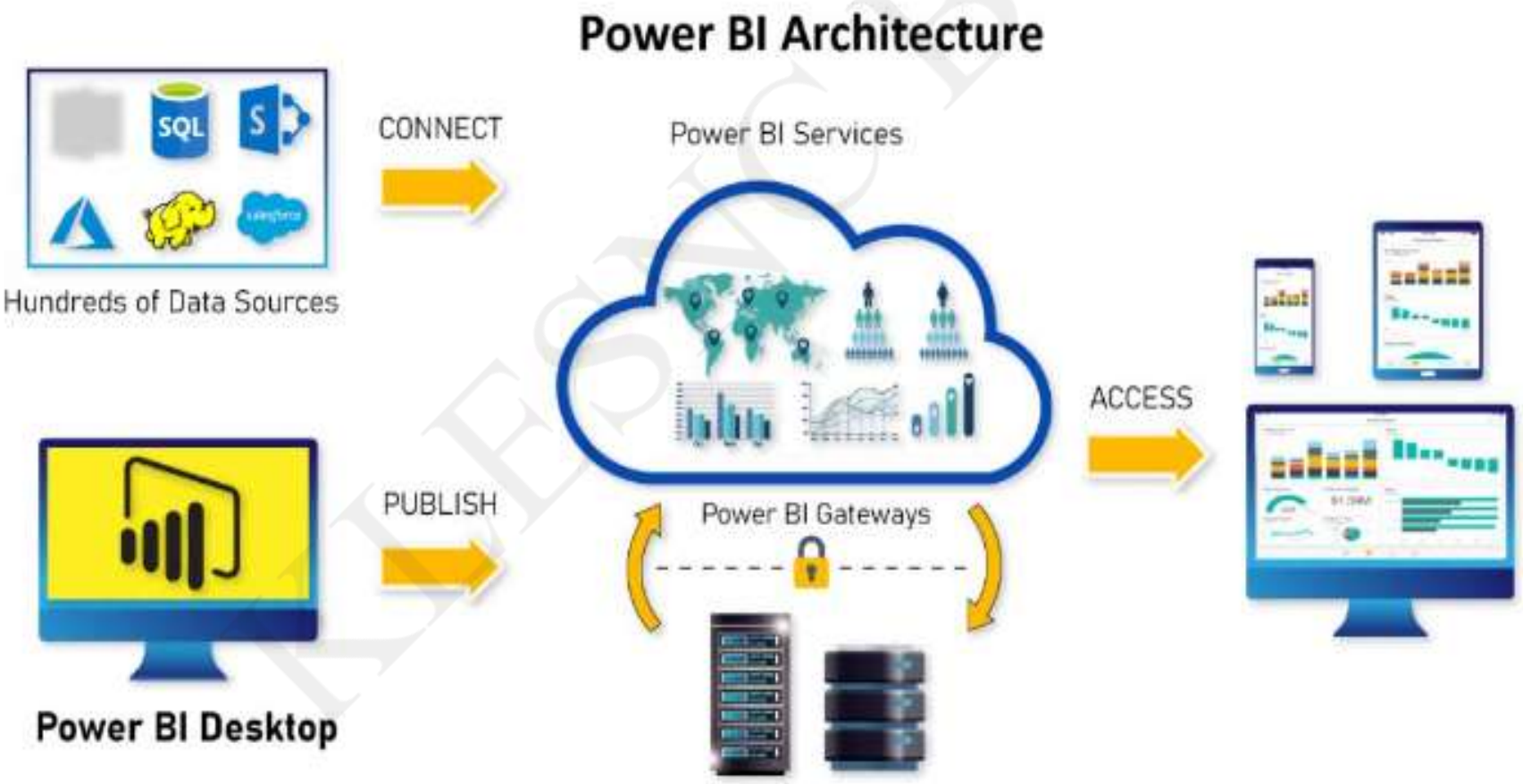


1. **Power Query:** It is used to access, search, and transform public and internal data sources.
2. **Power Pivot:** power pivot is used in data modelling for in-memory analytics.
3. **Power View:** By using the power view, you can analyse, visualize, and display the data as an interactive data visualization.
4. **Power Map:** It brings the data to life with interactive geographical visualisation.
5. **Power BI Service:** You can share workbook and data views which are restored from on-premises and cloud-based data sources.
6. **Power BI Q&A:** You can ask any questions and get an immediate response with the natural language query.
7. **Data Management Gateway:** you get periodic data refreshers, expose tables, and view data feeds.
8. **Data Catalog:** By using the data catalog, you can quickly discover and resuse the queries.

KLESNC BCA

Power BI Architecture

Power BI Architecture



RIESW BCA

- **Power BI architecture is an Azure-based solution**. This system can connect to a variety of data sources. You can use Power BI Desktop to produce reports and data visualizations based on the dataset. To obtain continuous data for reporting and analytics, the Power BI gateway is linked to on-premise data sources.
- The cloud services are used to publish Power BI reports and data visualizations. You can connect to their data from anywhere by using Power BI mobile apps. **Apps for Power BI are available for Windows, iOS, and Android**.

KLE SMC BCA

1. Data Sourcing and Integration:

- The data is Extracted from different sources such as excel, spreadsheet, CSV files, and different Databases. They can come from different servers. The data from various sources can be in different types and formats.
- If you import the file into the Power BI, it compresses the data sets up to 1GB, and it uses a direct query if the compressed data sets exceed more than 1GB. Then the data is integrated into a standard format and stored at a place called a staging area. There are two choices for big data sets. They are as follows: Azure Analytics Services and Power BI premium.

2. Data Transforming:

- Integrated data is not ready to visualize data because the data should be transformed. To transform the data, it should be cleaned or pre-processed. For example, redundant or missing values are removed from the data sets. After data is pre-processed or cleaned, business rules are applied to transform the data. After processing the data, it is loaded into the data warehouse.

3. **Reporting & Publishing:**

After sourcing and cleaning the data, you can create the reports. Reports are the visualization of the data in the form of slicers, graphs, and charts. Power BI offers a lot of custom visualization to create the reports. After creating reports, you can publish them to power bi services and also publish them to an on-premise power bi server.

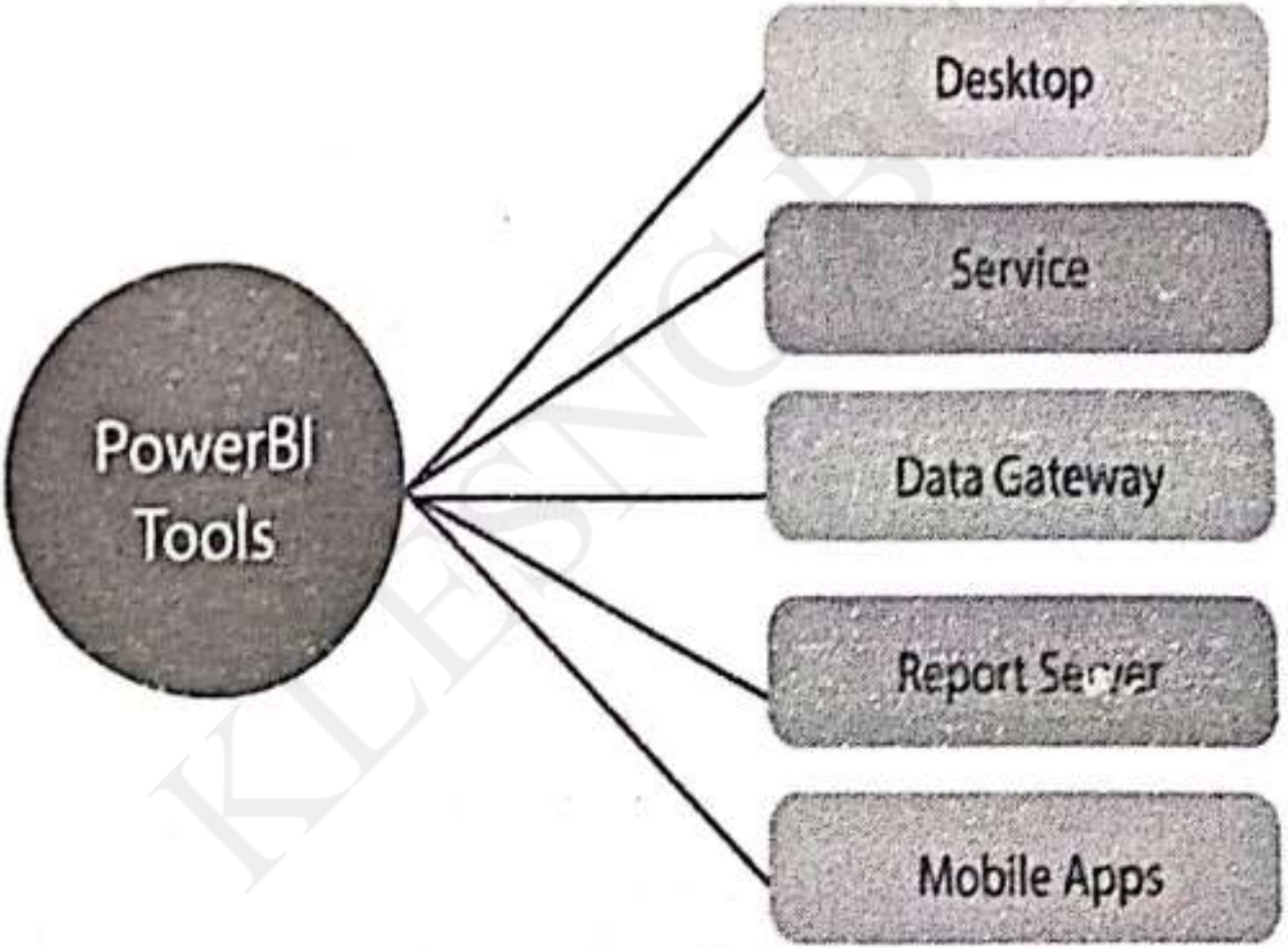
4. **Creating Dashboards:**

You can create dashboards after publishing reports to Power BI services, by holding the individual elements. The visual retains the filter when the report is holding the individual elements to save the report. Pinning the live report page allows the dashboard users to interact with the visual by selecting slicers and filters.

Note: Power BI service is the Software as a Service (SaaS) part of Power Bl. It is also known as Power BI Online. To access Power BI Service, you need to log in to Power BI service.

KLESMC BCA

Power BI Tools



1.Power BI Desktop:

- Power BI Desktop is a free application from Microsoft that allows users to connect to, transform, and visualize data. It is a key component of Microsoft Business Intelligence and allows users to import, analyze, and create reports.
- It is a primary authority and publishing tool. Power BI users and developers use it to create brand new models and reports.
- Power BI desktop tool is available at free of cost.

2. Power BI Service:

- The Power BI service is a cloud-based service, or software as a service (SaaS).The power BI data modules, dashboards, and reports are hosted in the online software as a service (SaaS). Sharing, administration, and collaboration happen in the cloud. It supports report editing and collaboration for teams and organizations.You can connect to data sources in the Power BI service, too, but modeling is limited.
- Power BI service is available at the pro license, and the user has to pay $10 per month.

KUEN CBCA

3. Power BI data gateway: It works as the bridge between the Power BI service on-premises data sources such as Import, Direct Query, and Live Query. BI Admin installs Power BI. Gateway act as a gatekeeper for the on-premises data source. If anyone needs to access on-premises data from a cloud or web- based app, the request goes through the gateway.

4. Power BI Report Server: It hosts paging reports, mobile reports, KPIs, and Power BI desktop reports. It requires updates in every four months and managed by the IT team.

5. Power BI Mobile Apps: It is available for Android, iOS, and windows. Microsoft intune manages it by using this tool. You can view reports and dashboards on the Power BI Service Report Server.

KLE SNC BCA

Some Basic Concepts :

- Dashboard is a collection which contains zero or more tiles and widgets. Each tile displays a single visualization that was created from a dataset and pinned to the dashboard. Each dashboard is used to represent a customized view of some subset of the underlying datasets.
- **Reports:-**A Power BI report is one or more pages of visualizations (charts, graphs and images).Reports can be created from scratch, imported to dashboard and created using datasets.
- **Datasets:-**A dataset is something that you import or connect to. Datasets can be refreshed, renamed, explored and removed. Each listed dataset represents a single source of data.

KLESNC BCA

Power BI Data Access :

In Power BI, data access refers to the ability to connect to and retrieve data from various sources for the purpose of creating reports and dashboards. Power BI provides a wide range of options for data access to accommodate different types of data sources.

1. Data Sources

Files

- Excel (.xlsx, xlxm)-a workbook can have data entered manually or data which is queried and loaded from external data sources. Data can be in simple worksheets or loaded into a data model
- Comma Separated Value (.csv) - Files are simple text files with rows of data. Each row can contain one or more values, each separated by a comma
- Power BI Desktop (.pbi) - You can use Power BI Desktop to query and load data from external data sources, extend your data model with measures and relationships, and create reports. You can import your Power BI Desktop file into your Power BI site.

KLE BCA

Databases

- Databases in the Cloud – From the Power BI service, you can connect live to Azure SQL Database, Azure SQL Data Warehouse ,etc. Connections from Power BI to these databases are live.
- Databases on-premises – From the Power BI service, you can connect directly to SQL Server Analysis Services Tabular model databases. A Power BI Enterprise Gateway is required.

Online services

- It including web services, APIs, and online platforms like SharePoint, Dynamics 365, Salesforce, and more.

KLESNC BCA

2. Data Connections

Power BI offers different methods to connect to data sources, such as:

- **Direct Query:** This allows Power BI to send queries directly to the data source in real-time, without importing the data into Power BI.
- **Import:** Data can be imported into Power BI, and reports are created based on the imported data.

3. Power Query Editor

- Power Query is a powerful data transformation tool within Power BI that enables users to connect to various data sources, transform and shape the data, and then load it into the Power BI model.
- It supports cleaning, transforming, and merging data from different sources before loading it into the Power BI model.

KUESTIC BCA

4. Gateways:

- Power BI Gateway is a tool that allows for secure and efficient communication between Power BI and on-premises data sources.

5.Data Refresh:

- For datasets that are regularly updated, Power BI provides data refresh options to keep the data in reports up-to-date. The refresh can be scheduled for automatic updates.

KLESNC BCA

Visualization Techniques for Spatial Data, Geospatial Data, Time-Oriented Data, Multivariate Data:

- Spatial DATA: Spatial data, refers to data that includes information about specific locations on the Earth's surface.
- A pair of latitude and longitude coordinates defines a specific location on earth.

EX: locations of various cities around the world

- Spatial data are of two types according to the storing technique, namely, raster data and vector data.
- Raster data are composed of grid cells identified by row and column. The whole geographic area is divided into groups of individual cells, which represent an image. Satellite images, photographs, scanned images, etc., are examples of raster data.
- Vector data are composed of points, polylines, and polygons. Wells, houses, etc., are represented by points. Roads, rivers, streams, etc., are represented by polylines. Villages and towns are represented by polygons.

KUESNC BCA

Geospatial Data:

- Geospatial data, or geodata, is data that includes information related to locations on the Earth's surface. You can map objects, events, and other real-world phenomena to a specific geographical area identified by latitude and longitude coordinates.
- Geospatial data is a subset of spatial data and is specifically related to the Earth's surface and its features. It involves the use of geographic information systems (GIS) to analyze and interpret spatial data.
- Examples of geospatial data include weather maps, traffic and accident data. This information has a geographic component that can tie it to an address or relative location.
- EX: Additional information such as the population, area, and elevation of each city.

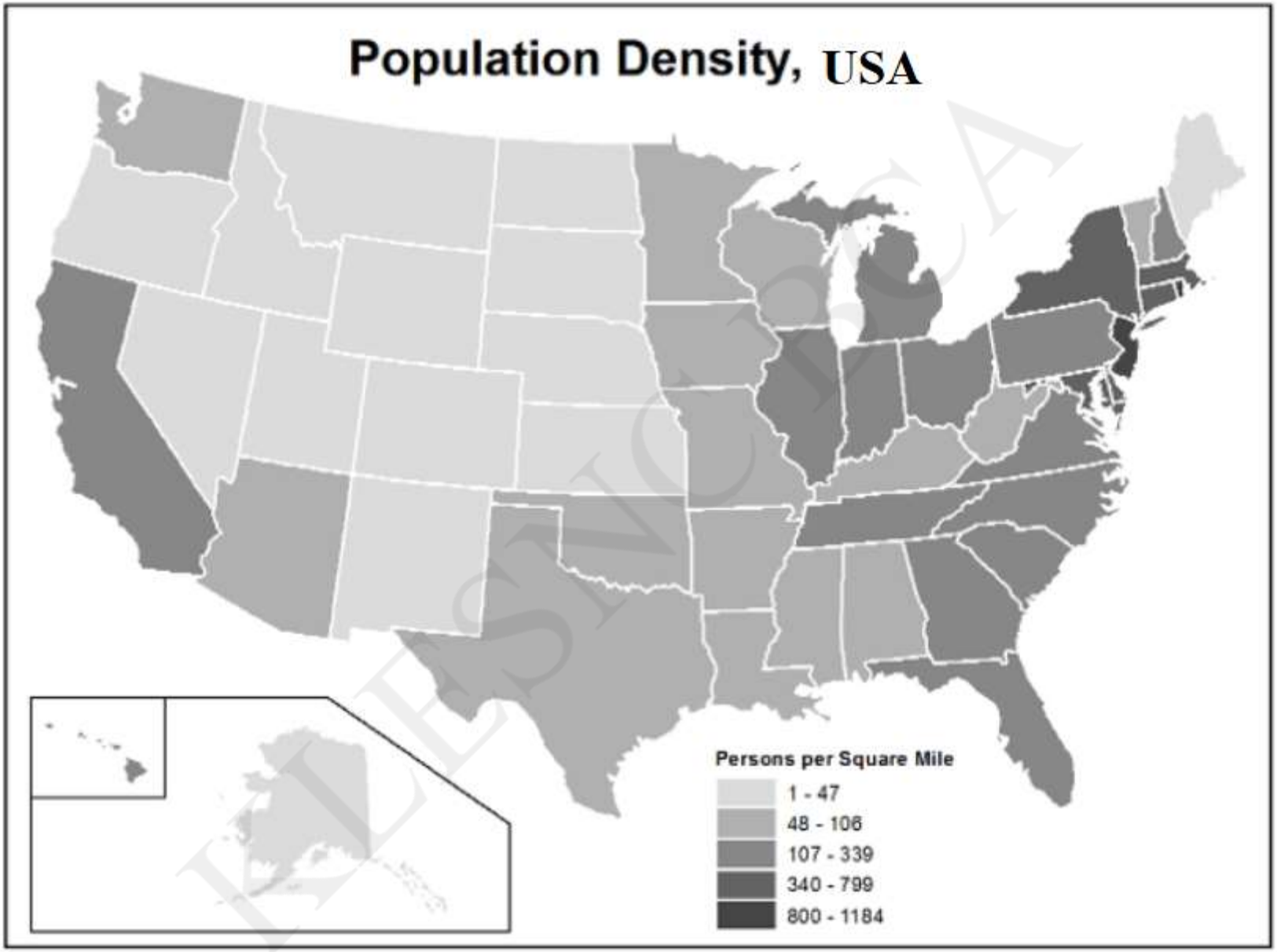
KLESNC BCA

- **Visualization Techniques for Spatial Data, Geospatial Data**

1. Choropleth Maps:

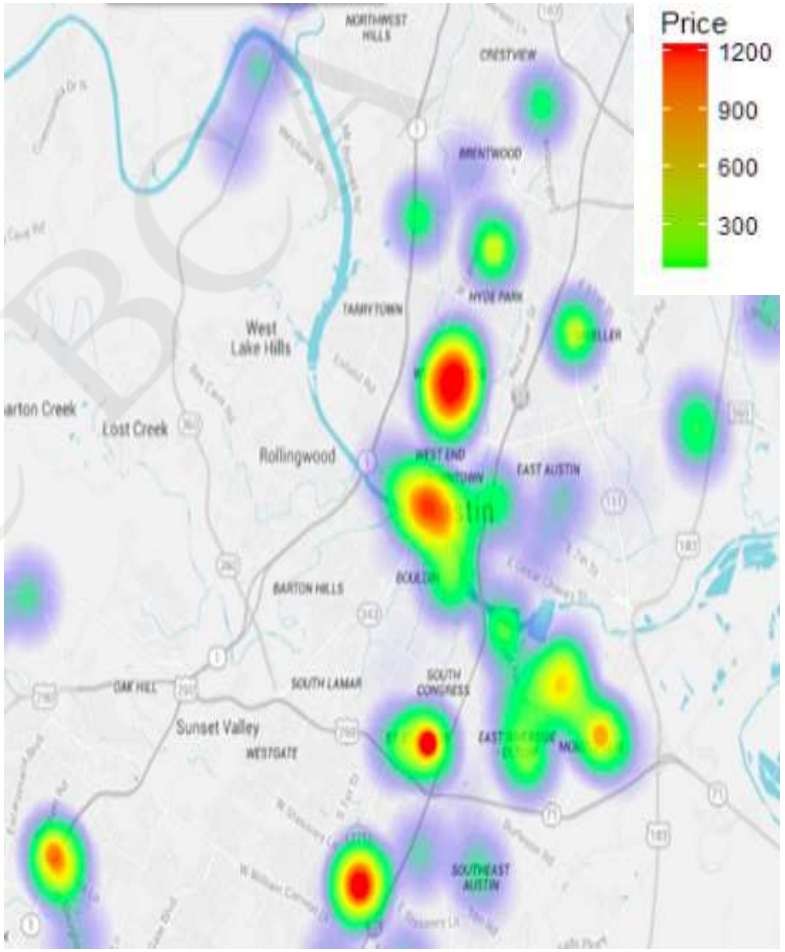
- Choropleth map uses different colorings, color shades, or pattern fills in various areas of a map to represent the value or range of values of some variable. Color shades are patterned differently for the represented geographic areas. The shaded areas are patterned according to statistics and values that are displayed on the map.
- Ex: Population density in USA , here the states with lower population density are shaded with lighter Gray Color and Higher population density are shaded with Dark gray color. The states with population densities are between two extremes are shaded on a continuum from the lightest Gray to darkest gray.

KBECNOBCA


Population Density, USA
Persons per Square Mile
1 - 47
48 - 106
107 - 339
340 - 799
800 - 1184

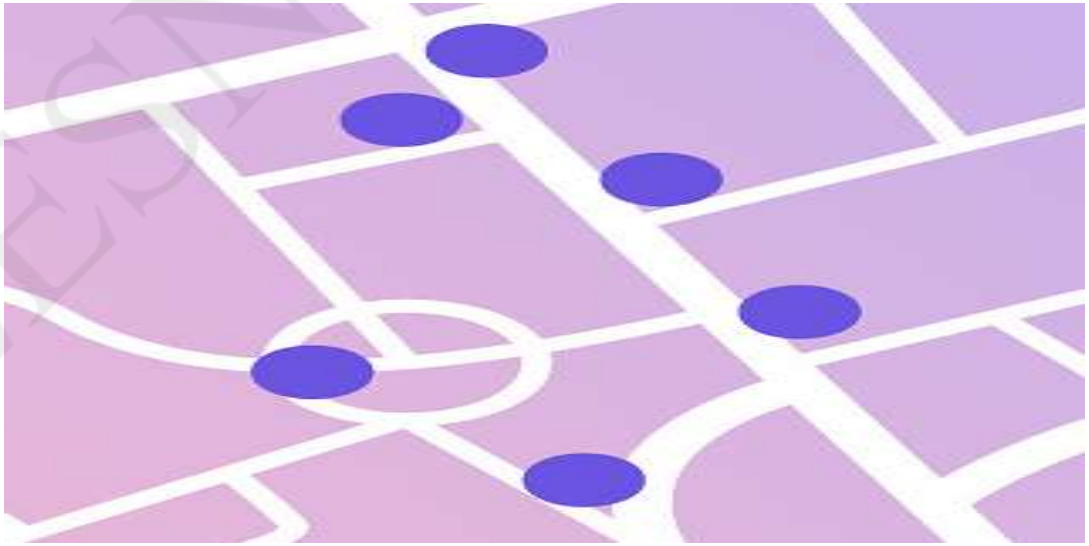
2. Heat maps :

- Heat maps use colors to communicate numeric data by varying the underlying values. Heat maps can be used to represent large sets of spatial data. Data values are represented red, green, and blue (RGB). In this way, the color variations represent continuous changes in the values on the map.
- The main purpose of heatmap visualization is to uncover patterns, trends, and relationships within the data by highlighting areas of high or low values, and lowest value in the heat map to dark blue, the highest value to a bright red.



Dot map :

- Dot maps, alternatively known as point maps, use dots to represent the data. Dots are placed into an approximate location where the data value was obtained. In general, dots are equally sized and are placed based on the geographical location. The user can obtain information by hovering around the dot on the map.



KLE SNC BCA

Bubble map :

- In a bubble map, a data source is properly represented with a bubble (circle) of variable. The size of the bubble represents the number of cases at a particular location. Bubbles are distributed over the map according to their geographical coordinates. Bubble maps are simple and easy to read.
- bubble maps can show several pieces of information in a single view by using a different color for each type of information.

example of a bubble map representing sales data for different product categories in various regions. In this example, we'll use different colors to represent different product categories, and the size of each bubble will indicate the sales volume for that category in a specific region.



Topographic map :

- A topographic map refers to a detailed, graphical, and accurate representation of features that appear on the Earth's surface.
- A topographic map is a map that represents the locations of geographical features. geographical features can be mountains, valleys, plain surfaces, water bodies and many more.



Flow map :

- Flow maps, also known as 'path' maps, are more specialized versions of line maps. Instead of focusing on physical features of the earth, they are used to represent the movement of things across the earth over time. Flow maps depict the movement of people, goods, or information between different locations.
- A map with a path typically includes a line or series of lines that represent the path or route

Cartograms:

- Cartograms distort the geographic boundaries of regions based on a specific variable (population, GDP, etc.).

Ex: Population Cartogram:

- Regions are resized based on their population density or total population. More populated areas appear larger, while less populated areas shrink.

Proportional Symbol Maps:

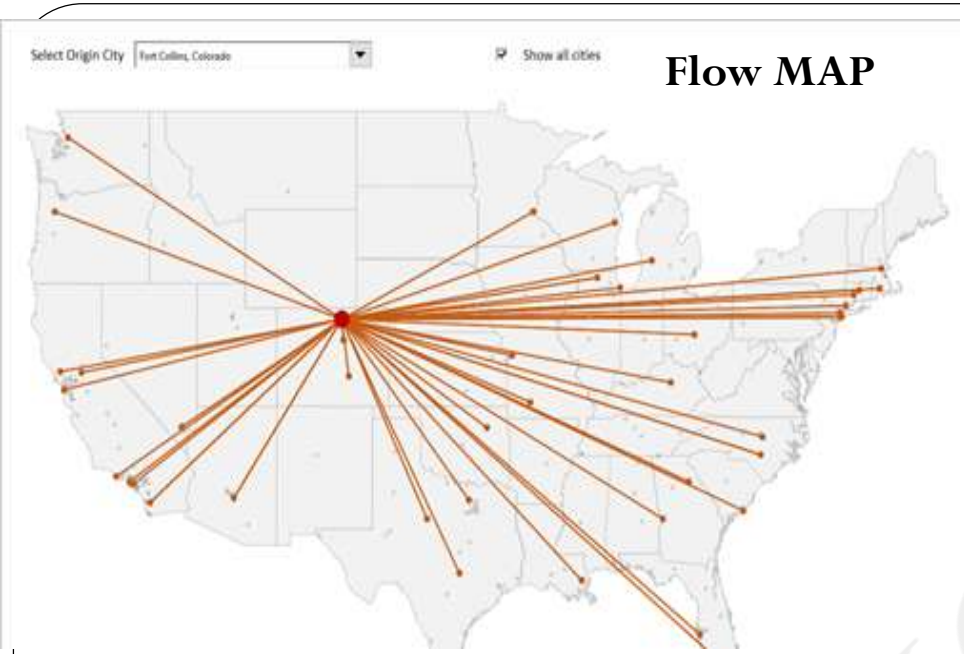
- Proportional symbol maps use symbols of different sizes to represent the quantity of a variable at specific locations.
- The size of the symbol corresponds to the magnitude of the variable.

Ex : Sales data

- Symbols represent the sales volume or revenue of different retail locations. Larger symbols may indicate stores with higher sales, while smaller symbols represent locations with lower sales.

KUFSIC BCA

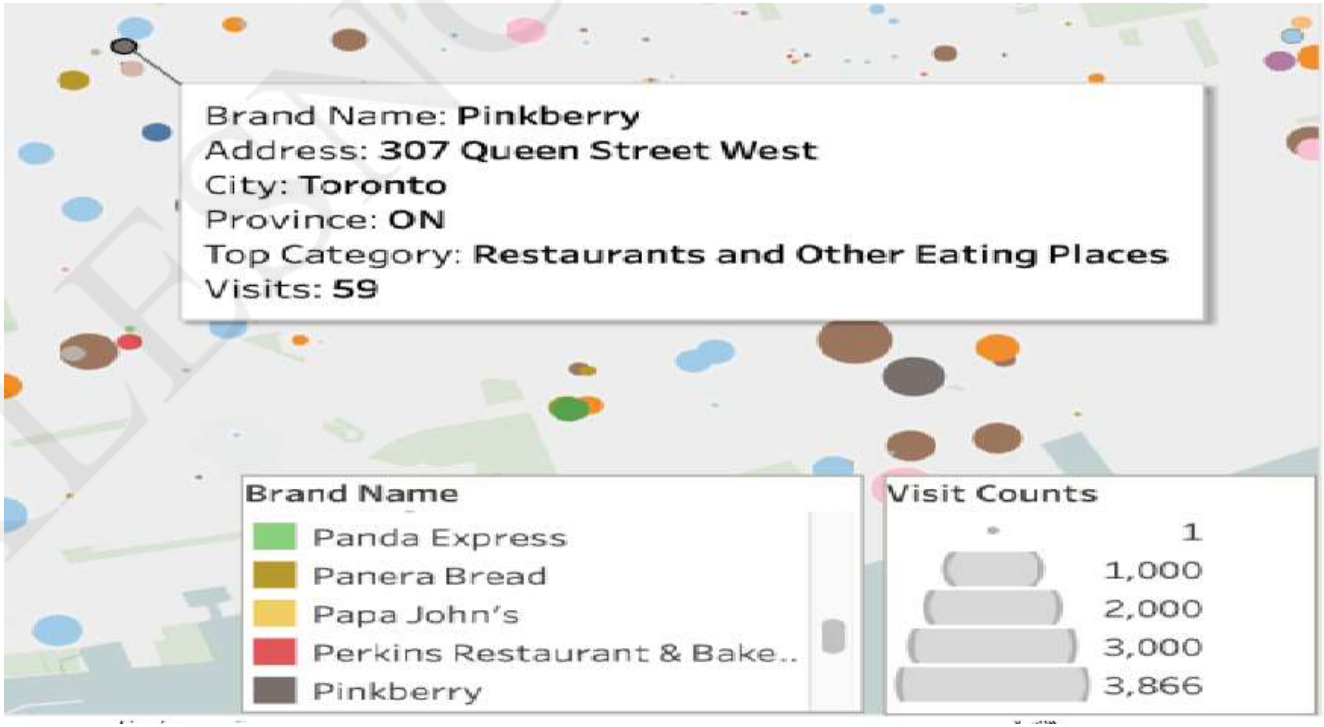
Flow MAP





Cartograms MAP:

Proportional Symbol Maps:



Visualization Techniques for Multivariate Data:

Multivariate Data:

- Multivariate data contains, at each sample point, multiple scalar values that represent different simulated or measured quantities.
- Multivariate analysis takes place when you have a data set with 4 or more dependent variables which are to be examined against an independent variable or variables.

1. **Geometric Representations :**

- Geometric representations refer to visual depictions or graphical presentations of data using geometric shapes, patterns, or structures.

KLE CET BCA

Parallel Coordinates:.

- A method for visualizing multivariate data in a high-dimensional space. Each variable is represented by a vertical axis, and lines connect points across axes, providing insights into patterns and clusters. parallel coordinates plots are effective for displaying relationships between multiple numerical variables on parallel axes.

Ex: Financial Analysis:

- Imagine a dataset with financial indicators such as revenue, expenses, profit margin, and stock price. Parallel coordinates can be used to display these financial metrics for different time periods, helping analysts identify trends and correlations.

2. Scatterplots Matrix:

- A scatterplot matrix consists of multiple scatterplots arranged in a grid, with each pair of variables represented in a separate plot.
- It helps visualize the relationships between all pairs of variables simultaneously.

Ex: **Exam Scores vs. Study Hours:**

- Scatter plots can be used to examine the correlation between study hours and exam scores. Each point represents a student, with study hours on one axis and exam scores on the other, helping to identify if more study time leads to higher scores.

KLESNCBCA

3. Icon Representations :

- The concept of an icon represent each independent variable with a number of attributes that can vary to represent the dependent variables.
- There are two common icon representation techniques star plots and Chernoff faces.

3.1Star plots :

- Radar charts (also known as spider or star plots) represent each variable as a spoke radiating from a central point.
- Multiple observations are connected to create a polygon, allowing for visual comparisons.

Ex: **Performance Evaluation:**

- Employee performance across different metrics, such as communication skills, teamwork, productivity, and leadership, can be visualized using a star plot. Each axis represents a different skill or attribute.

KLEBSC BCA

3.2 Chernoff faces :

- Chernoff Faces use facial features to represent multiple variables. Each variable is associated with a specific facial feature (e.g., eyes, mouth, etc.).
- Changes in facial features can convey patterns in the data.

Ex: Size of eyes can represent a categorical variable, such as gender (smaller for females, larger for males). Thickness and angle of eyebrows can represent a numeric variable, like intelligence or education level.

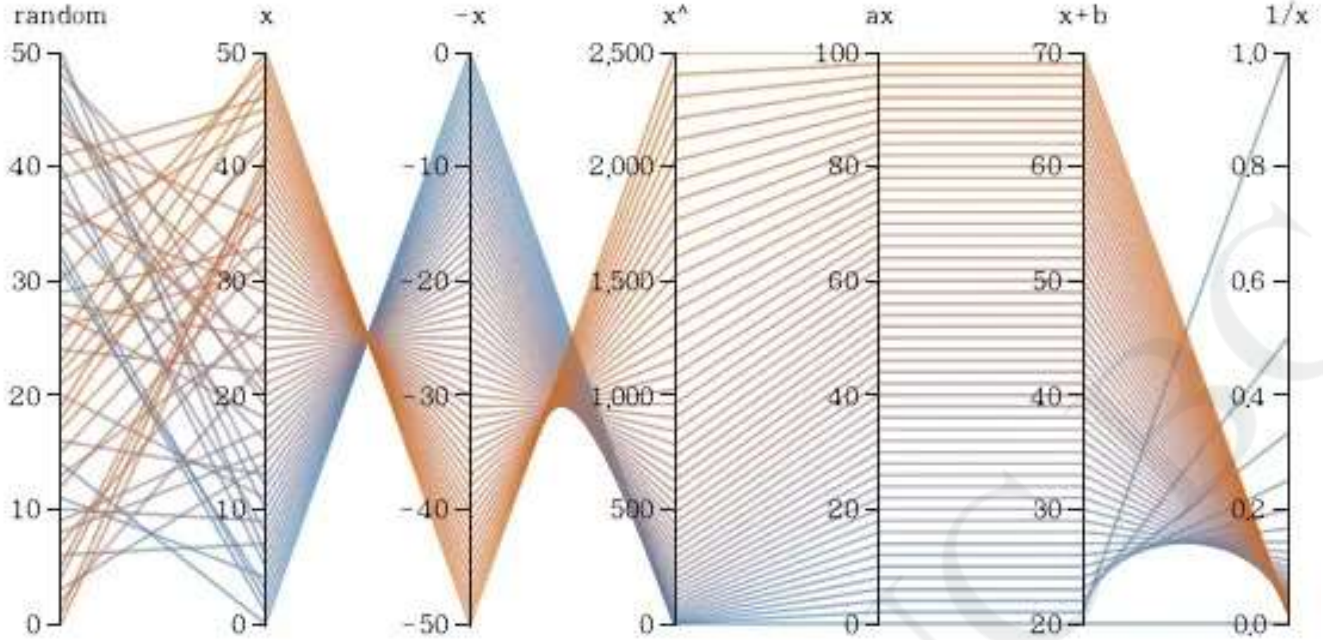
4. Pixel-Oriented Representations

- This form of representation uses the idea that the pixel is the smallest unit available on a screen and thus represents each data unit in pixels typically forming an image.

Ex: Image processing

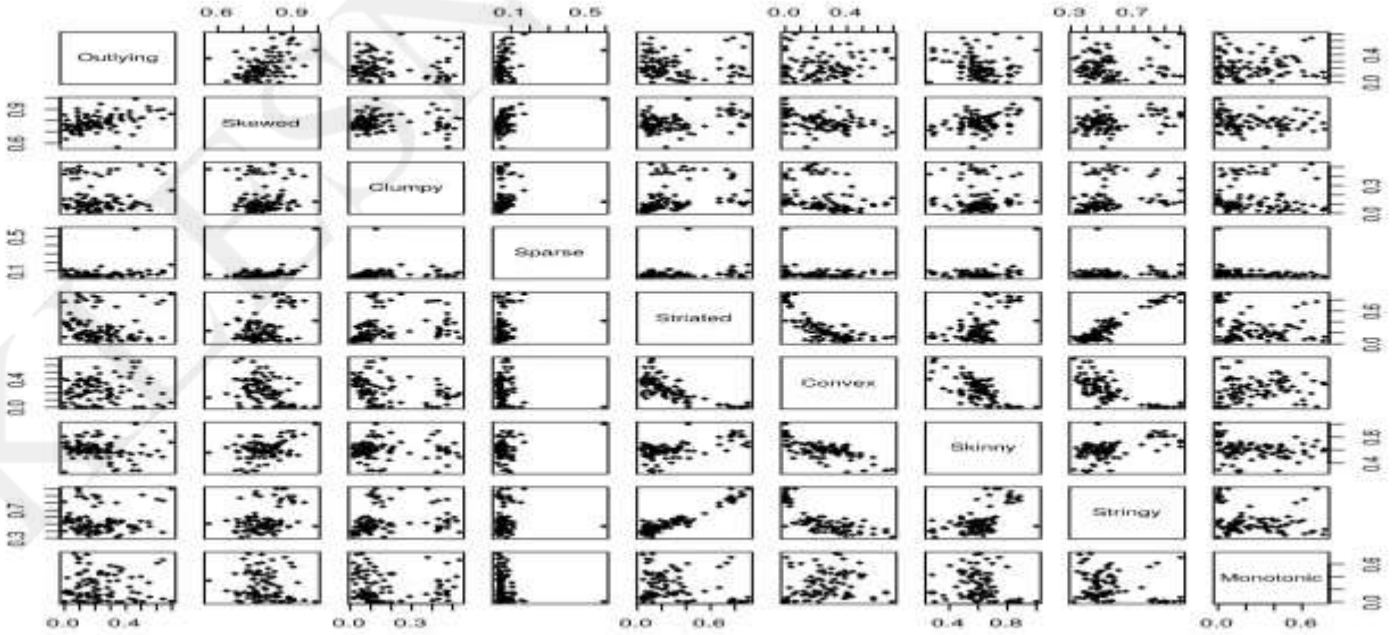
- For example, applying filters, adjusting brightness, or changing color balance involves operations on individual pixels.

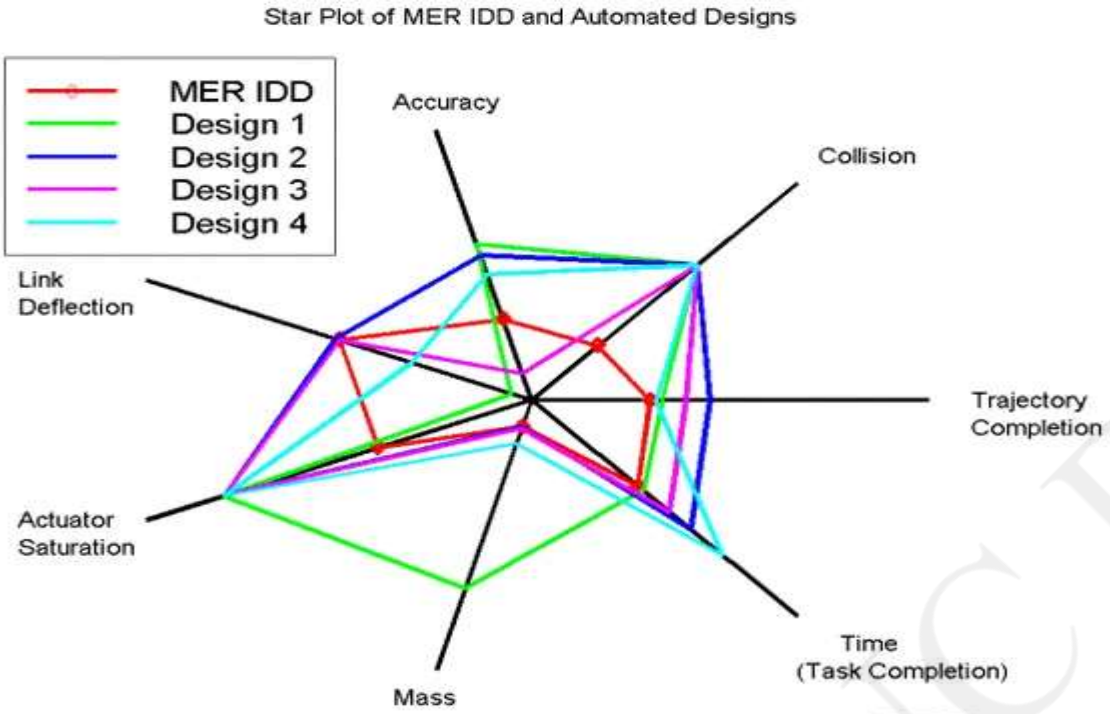
KLESNCBCA



Parallel Coordinates:.

Scatterplots Matrix:





Star PLOT

Chernoff faces :



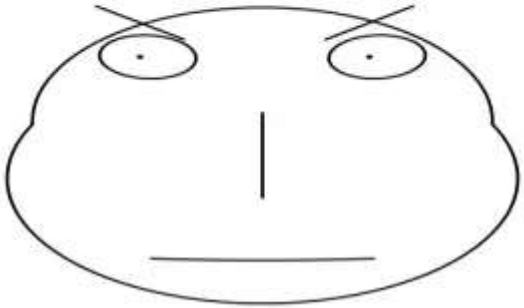
AARONSON,L.H.



ALEXANDER,J.M.

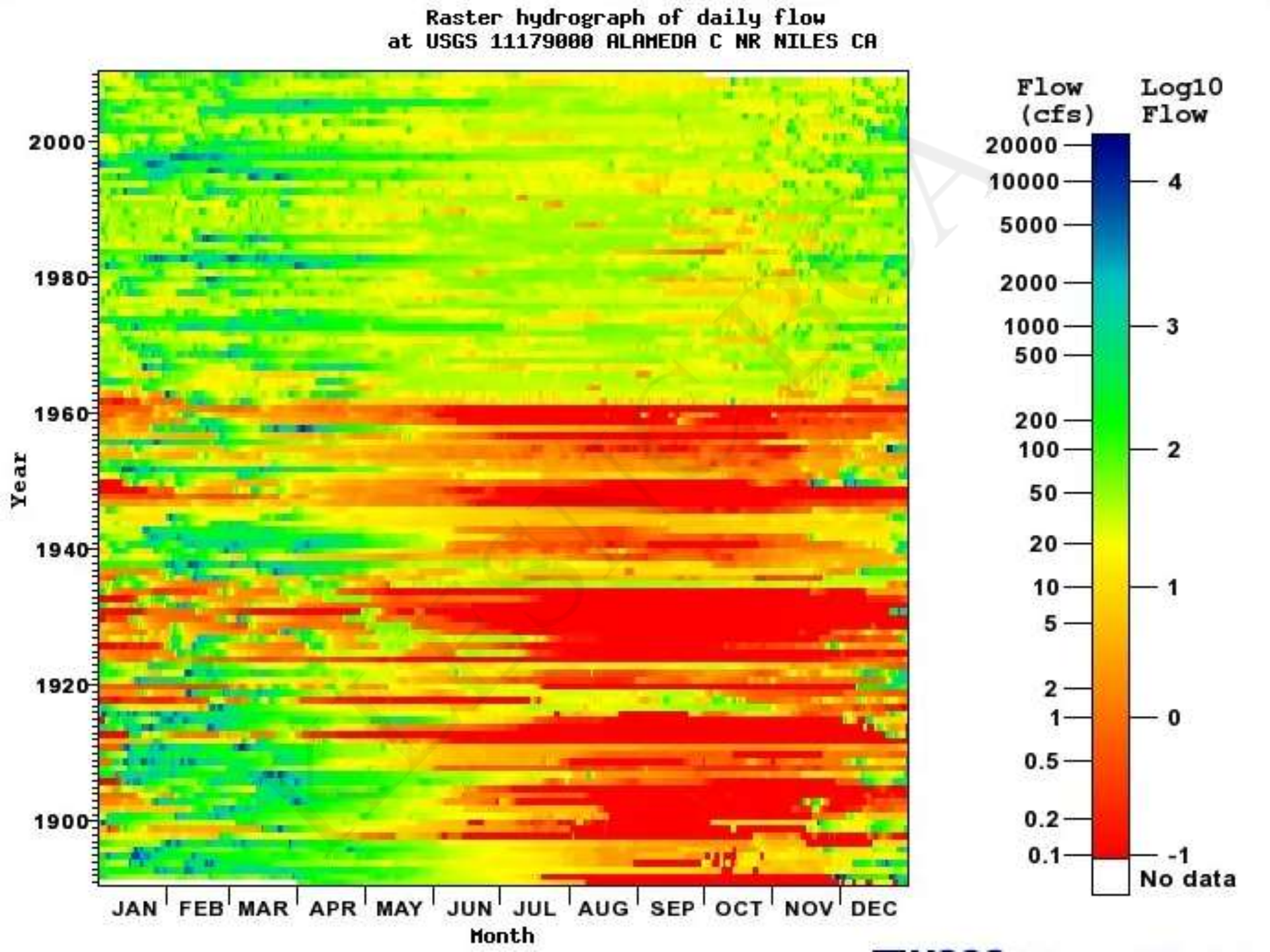


BRACKEN,J.J.



BURNS,E.B.

KLESNC BCA


Raster hydrograph of daily flow
at USGS 11179000 ALAMEDA C NR NILES CA
Year
2000
1980
1960
1940
1920
1900
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
DEC
Month
Flow (cfs)
Log10 Flow
20000
10000
5000
2000
1000
500
200
100
50
20
10
5
2
1
0.5
0.2
0.1
4
3
2
1
0
-1
No data

- **Visualization Techniques for Time -oriented Data:**

Time -oriented Data: It is a sequence of data points that occur in successive order over some period of time.

Bar charts:

- Bar charts are useful for comparing data across categories, where the length of each rectangular bar represents the magnitude of a particular data point.
- Ex: sales performance of different products.

Line Charts :

- Line Charts are used to visualize trends over time, with a series of points connected by lines where each point represents a data point for a particular period.
- **X-Axis:** Represents the independent variable or the domain (e.g., time, distance, categories).
- **Y-Axis:** Represents the dependent variable or the range (e.g., values, quantities).

Ex: Illustrating the trend of monthly sales over a year.

KLESIC BCA

Scatter plots

- Scatter plots, on the other hand, are used to visualize the relationship between two variables. A series of points represents a pair of values for two variables.

Ex: Demonstrating a positive correlation between the number of hours studied and exam scores.

Heat maps

- Heat maps are helpful in visualizing the distribution of values in a matrix or table. The colour of each cell in a coloured grid represents the magnitude of a particular data point.
- **Ex: Temperature Heatmap**

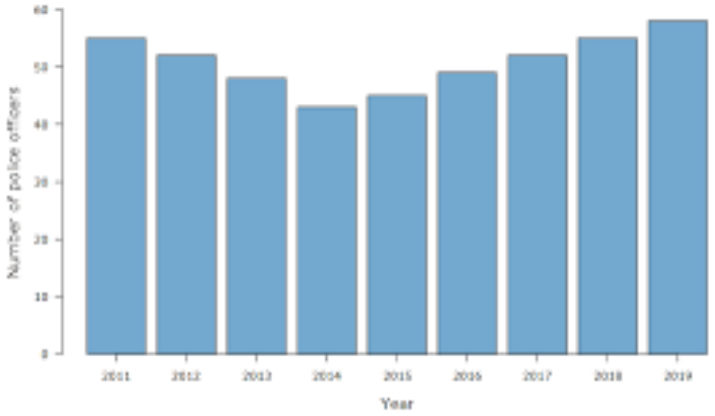
Area Charts:

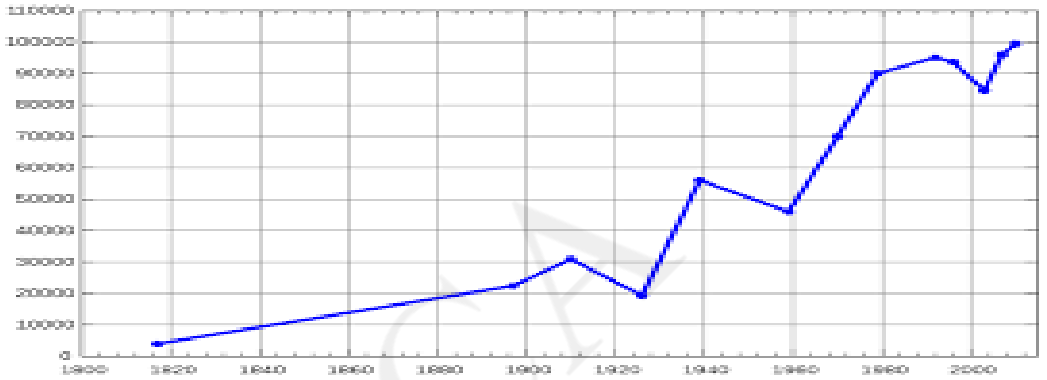
- An area chart is a type of chart that displays quantitative data over a continuous interval or time span.
- Area charts are similar to line charts but fill the area below the line, providing a visual representation of the cumulative values over time.
- Useful for comparing multiple time series on the same chart.

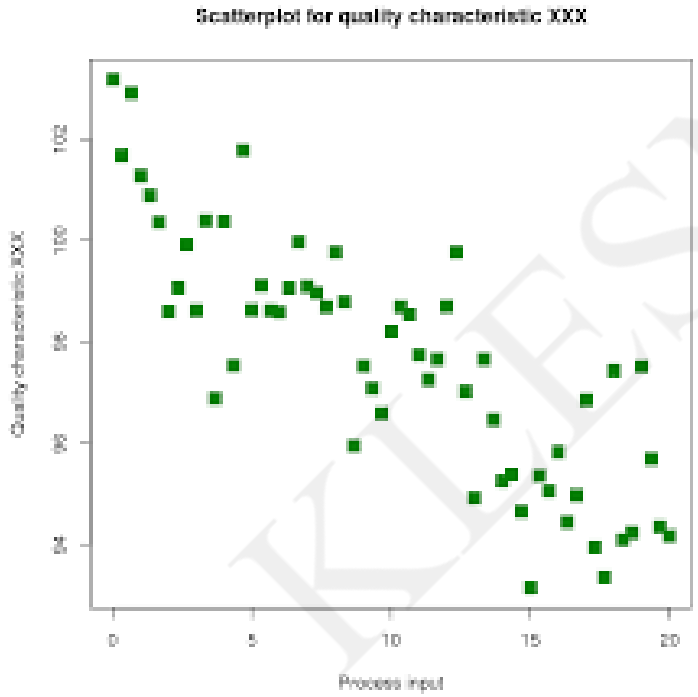
Ex: Quarterly Sales Performance

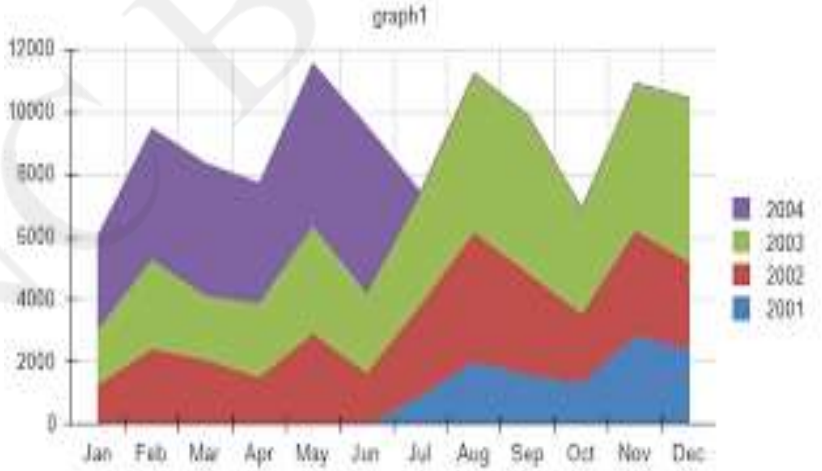
KTESNC BCA

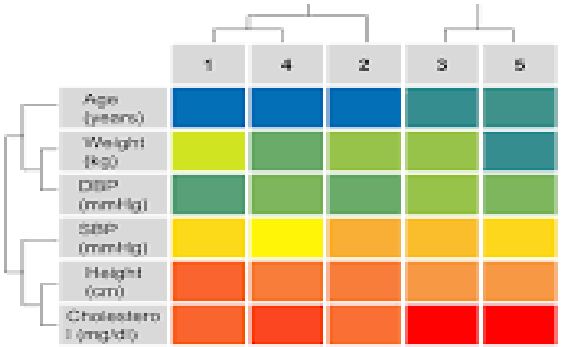
Chart 5.2.1
Number of police officers in Crimeville, 2011 to 2019











KUESNG BCA

Temperature Heatmap:

	January	February	March	April	May	June	July	August	September	October	November	December
City A	25°C	28°C	32°C	35°C	38°C	40°C	42°C	39°C	36°C	30°C	27°C	24°C
City B	18°C	21°C	26°C	28°C	32°C	35°C	37°C	36°C	33°C	28°C	22°C	19°C
City C	30°C	32°C	33°C	30°C	28°C	26°C	25°C	27°C	29°C	31°C	33°C	30°C

- Each cell corresponds to a specific month for a particular city.
- The color intensity in each cell represents the temperature value, with red indicating higher temperatures and blue indicating lower temperatures.

KLESNCBCA

Visualization Techniques for, Trees, Graphs, and Networks

KLE SCOE&A

Visualization Techniques for Trees :

- Visualization techniques play a crucial role in understanding and interpreting complex data structures such as trees, graphs, and networks.

Trees : A tree visualization displays hierarchical data with a collection of nodes (data points) and edges (hierarchical relations between nodes).

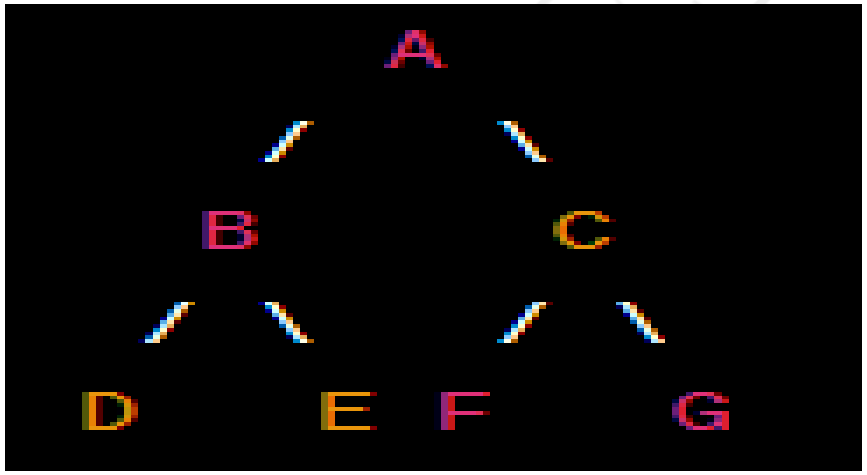
- A tree is defined as set of nodes and Edges.
- Every Edge has a pair of nodes: The parent node and the child node
- A child node has only one parent node
- A tree is a network of connected nodes where there are no loops
- Root node: The single that has no Parents
- Leaf node: The nodes that have no children
- Depth of the tree : Number of nodes from the root to the leaf.

KLESNC PCA

Common tree visualization techniques :

1. **Traditional Tree diagrams:**

- This involves using nodes and branches(Edges) to represent the hierarchical structure. Each node represents a decision or a concept, and branches(Edges) connect related nodes. Nodes can contain information such as data values, labels, or identifiers. The lines connecting nodes, indicating the relationships between them. An edge typically points from a parent node to its child node.



In this example, A is the root node, and B, C, D, E, F, and G are its children. The edges represent the relationships between the nodes.

2. **INDENTED LAYOUT:** Places all items along vertically Spaced rows.
Uses Indentation to show parent Child relationships

Example: Window explorer

Treemap :

•Treemaps display hierarchical data as a set of nested rectangles. Each level of the hierarchy is represented by a colored rectangle called a *branch* node. Each branch contains smaller rectangles called *leaf* nodes. Power BI uses the measure value to determine the rectangle size for branches and leaves.

•The rectangles are arranged by size with the largest branch node at the top left and the smallest branch at the bottom right. The arrangement of the leaf nodes within each branch follows the same order.

•Ex: disk space on a computer.

•Each folder or file is represented by a rectangle, and the size of the rectangle corresponds to the amount of disk space it occupies.

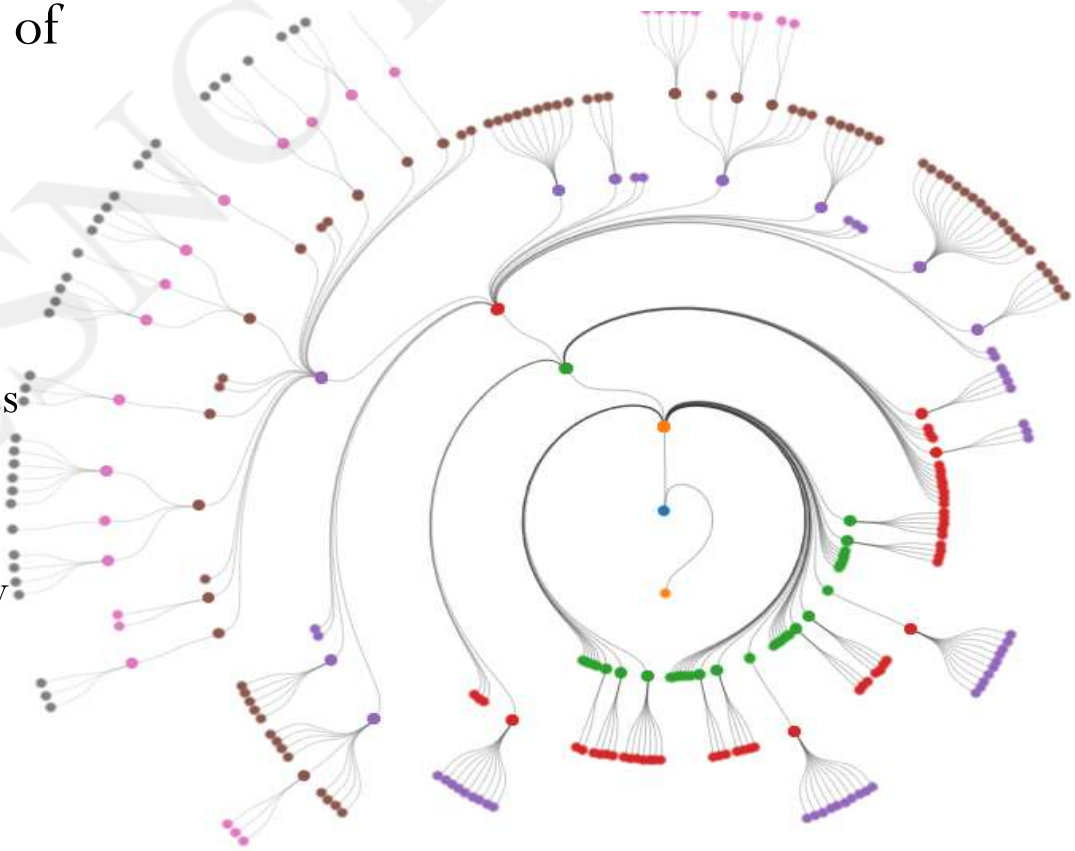
KLE SOC BCA

Radial tree :

A radial tree is a visual representation of a hierarchical structure where nodes are arranged in a circular layout. The root node is positioned at the center, and its child nodes radiate outward in a circular manner, forming levels of the hierarchy represented by concentric circles or arcs.



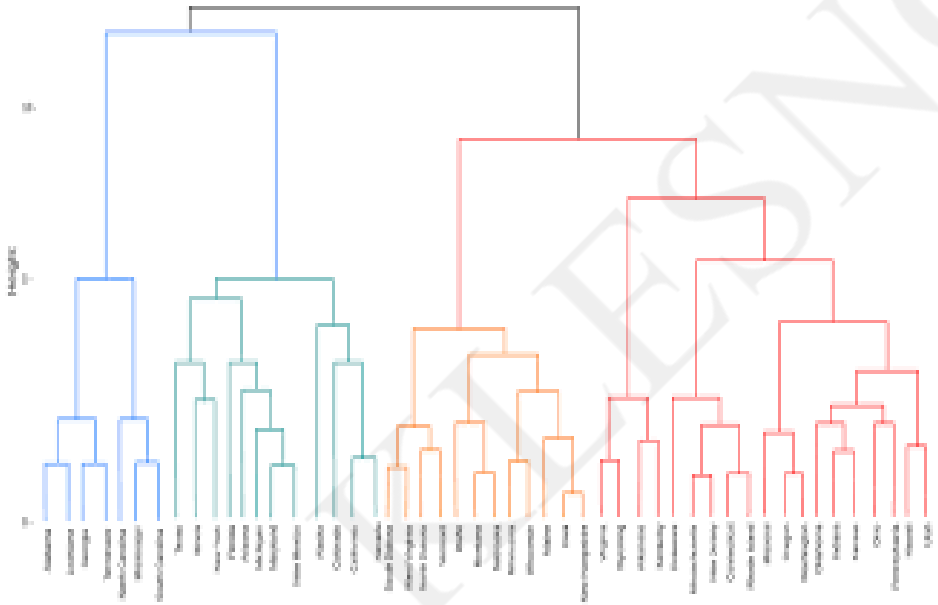
Radial tree



Ex: visualize organizational hierarchies
The central node represents the organization or a top-level executive, and branches extend outward to show the various levels of management and departments.

3. Dendrogram:

- Essentially a Layered Drawing with bended orthogonal edges.
- Layering are done according to the leaves. All the leaves are on the same layer.
- Ex: Commonly used in bioinformatics to represent the result of hierarchical clustering.



EX: Dendrograms can be applied in social network analysis to cluster individuals or groups based on social interactions, helping to identify communities or subgroups within a larger network.

RESNCBCA

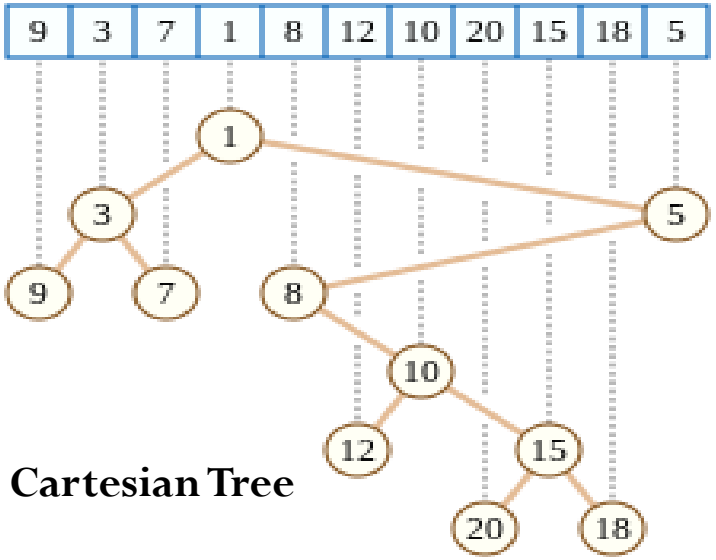
Mind Maps:

- A graphical representation of ideas, concepts, or tasks linked around a central theme. Useful for visualizing relationships and dependencies in a non-linear way.
- **EX:** Mind maps can be employed during meetings to capture and organize ideas, action items, and discussions. They serve as a visual record and aid in clarifying complex topics

Cartesian Tree:

- A binary tree representation where each node is associated with a point in a Cartesian coordinate system. Useful for visualizing binary tree structures.





Cartesian Tree

KLESIC BCA

Visualization Techniques for Graphs and Networks

- Graph visualization is a crucial aspect of understanding and analyzing complex relationships within data represented as graphs or networks.
- Network visualization, also known as graph visualization , is the process of visually presenting networks of connected entities as links and nodes.
- Graphs consist of nodes and edges, where nodes represent entities, and edges represent relationships between entities.
- Here are some visualization techniques for graphs and networks:

Matrix Plots:

- Represent the graph as a matrix, where rows and columns correspond to nodes, and cells indicate the presence or absence of edges. Particularly useful for adjacency matrices in sparse graphs.
- The matrix of plots shows possible positive relationships between all of the variables.
- Ex: A business analyst wants to study 22 successful small-to-medium size manufacturing companies. They collect number of clients, rate of return, and number of years in business.

KLE BCA

Arc Diagrams:

- An **arc diagram** is a special kind of network graph.
- Nodes are placed along a single axis, and arcs or curves connect nodes to represent edges.Suitable for directed graphs and displaying the flow of relationships.

Ex: Social Networks

- Arc diagrams can be used to visualize social networks, where individuals are represented as nodes, and relationships (friendships, collaborations) are depicted as arcs between them.

Chord Diagrams:

- Display relationships between nodes in a circular layout with chords connecting nodes. Effective for visualizing connections in a cyclic graph.

Ex:

- Chord diagrams can visualize relationships between different market segments, customer preferences, or product categories. This helps in identifying patterns in consumer behavior.

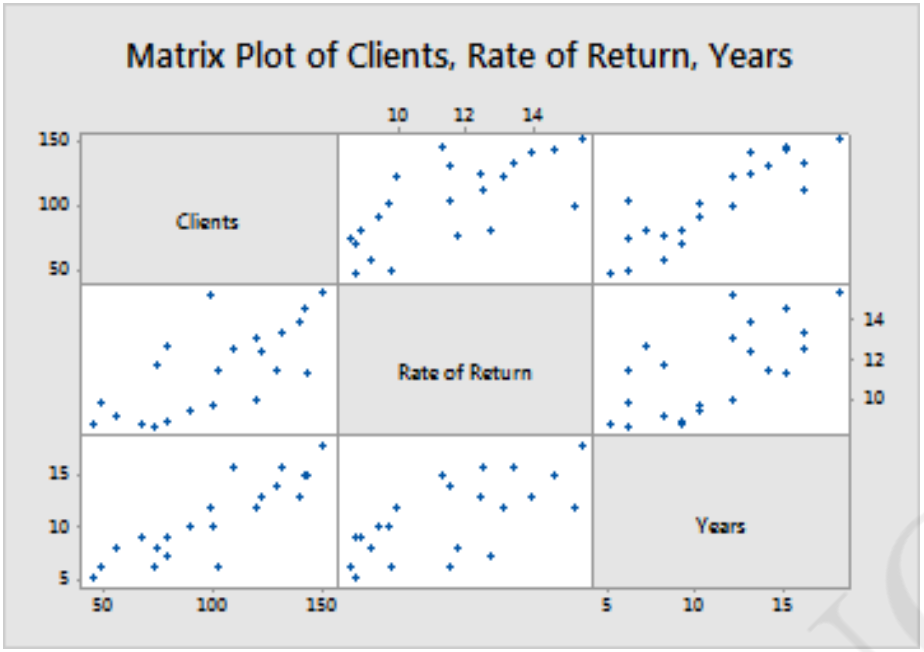
KLESNCBCA

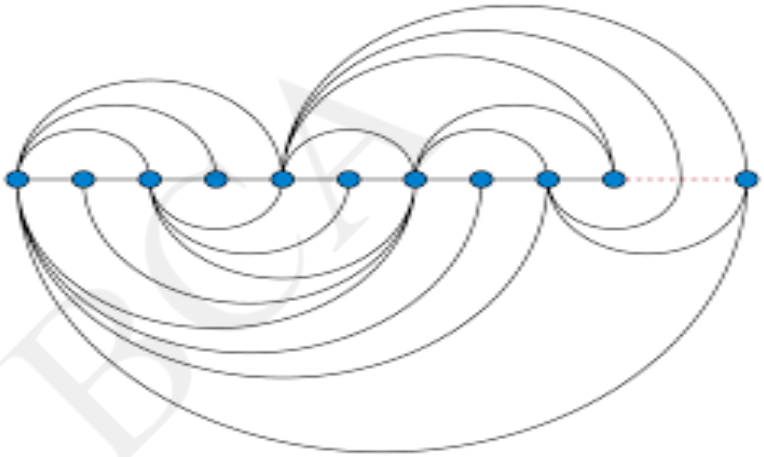
Animated Graphs:

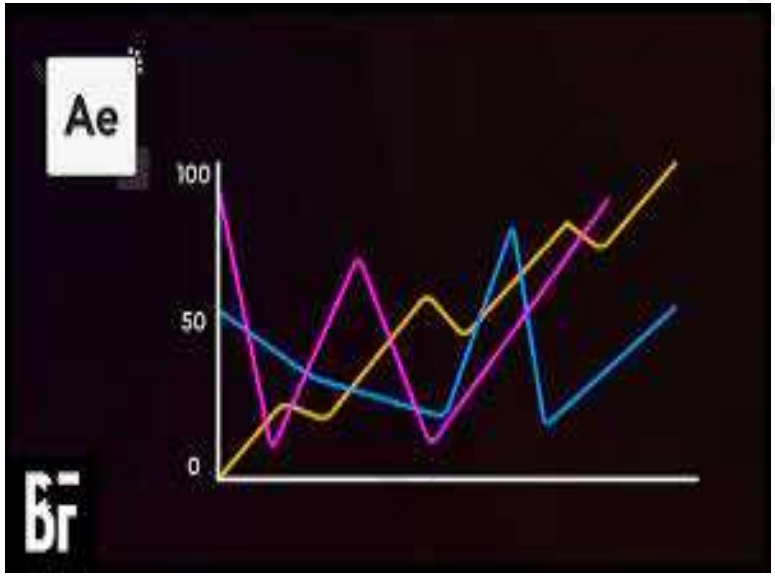
- Use animations to show the evolution of the graph over time or to highlight specific interactions. Helpful for dynamic or time-varying graph data.
- Animated graphs are excellent for visualizing changes in time series data.

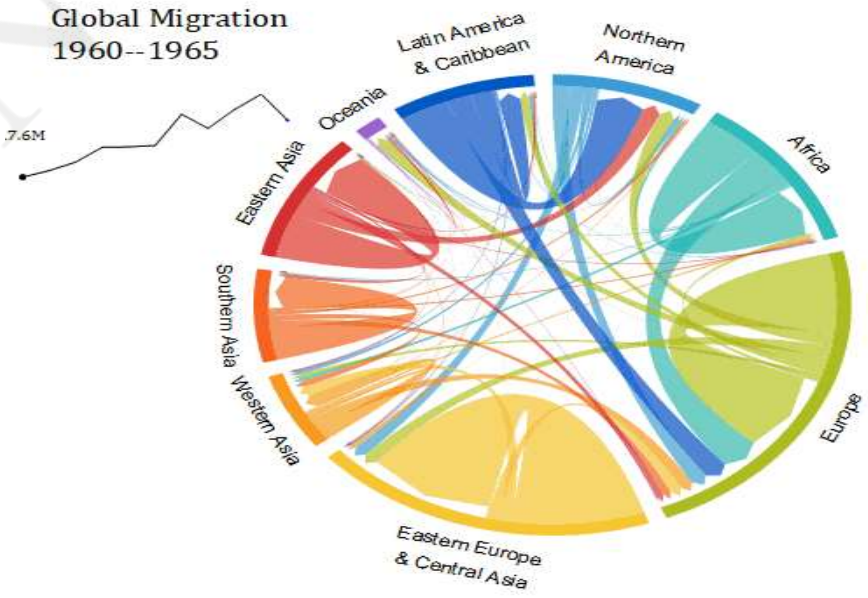
Ex: This can be useful in financial analysis, climate studies, or any domain where trends and patterns evolve over time.

KLESNC BCA


Matrix Plot of Clients, Rate of Return, Years
Clients
Rate of Return
Years
10
12
14
150
100
50
15
10
5
50
100
150
5
10
15
14
12
10




Ae
100
50
0
Br


Global Migration
1960--1965
7.6M
Latin America & Caribbean
Northern America
Oceania
Africa
Eastern Asia
Southern Asia
Western Asia
Europe
Eastern Europe & Central Asia

Text and Document Visualization

KLESNC BCA

Text and Document Visualization :

- Information contained in a Text document
- **Content:** Information Contained in the text itself
- **Structure:** Characterize how text is organized
- **Metadata:** Describes all types of information related to text that are not contained in the text itself.

Why visualize text?

- Understanding – get the "gist(main point)" of a document
- Grouping – cluster for overview or classification
- Compare – compare document collections, or inspect evolution of collection over time
- Correlate – compare patterns in text to those in other data, e.g., correlate with Social Network

KLESNCBCA

What is text data?

Documents

- Articles, books and novels
- E-mails, web pages, blogs
- Tags, comments
- Computer programs, logs

Collection of documents

- Messages (e-mail, blogs, tags, comments)
- Social networks (personal profiles)
- Academic collaborations (publications)

Text and document Visualization :

Word Clouds:

- A word cloud (also known as a tag cloud) is a visual representation of words. Cloud creators are used to highlight popular words and phrases based on frequency and relevance and displays the most used words in a text from small to large, according to how often each appears.

KLE CET BCA

- They provide you with quick and simple visual insights that can lead to more in-depth analyses.
- **Ex:** They give a glance into the most important keywords in news articles, social media posts, and customer reviews, among other text



- **Text Processing Pipeline**

1. Tokenization

- Segment text into terms.
- Remove stop words? a, an, the, of, to, be
- Numbers and symbols? #gocard, @stanfordfball, Beat Cal!!!!!!!!
- Entities? San Francisco, O'Connor, U.S.A.

2. Stemming

- Group together different forms of a word.
- Porter stemmer? visualization(s), visualize(s), visually visual
- Lemmatization? goes, went, gone go

3. Ordered list of terms

KLESNC BCA

Card Visuals:

- Utilize card visuals to display key metrics related to your text or document data. This could include the number of documents, average word count, sentiment score, etc.

Image and Web URL Fields:

- Power BI allows you to include images and web URL fields. You can use this feature to display document thumbnails, previews, or link to external documents or resources.

Map Visualizations:

- If your text data includes geographic information, use map visuals to display the geographical distribution of documents or entities.

Table and Matrix Visualizations:

- Use tables and matrices to display textual information. For example, you can create a table that shows document titles, authors, and other relevant details. Use conditional formatting to highlight important information.

KLE SNC BCA

Heatmaps for Text Data:

- Visualizing the intensity of word occurrences, sentiment scores, or other metrics in a document. Heatmaps provide a quick overview of patterns and trends.

Document Clustering:

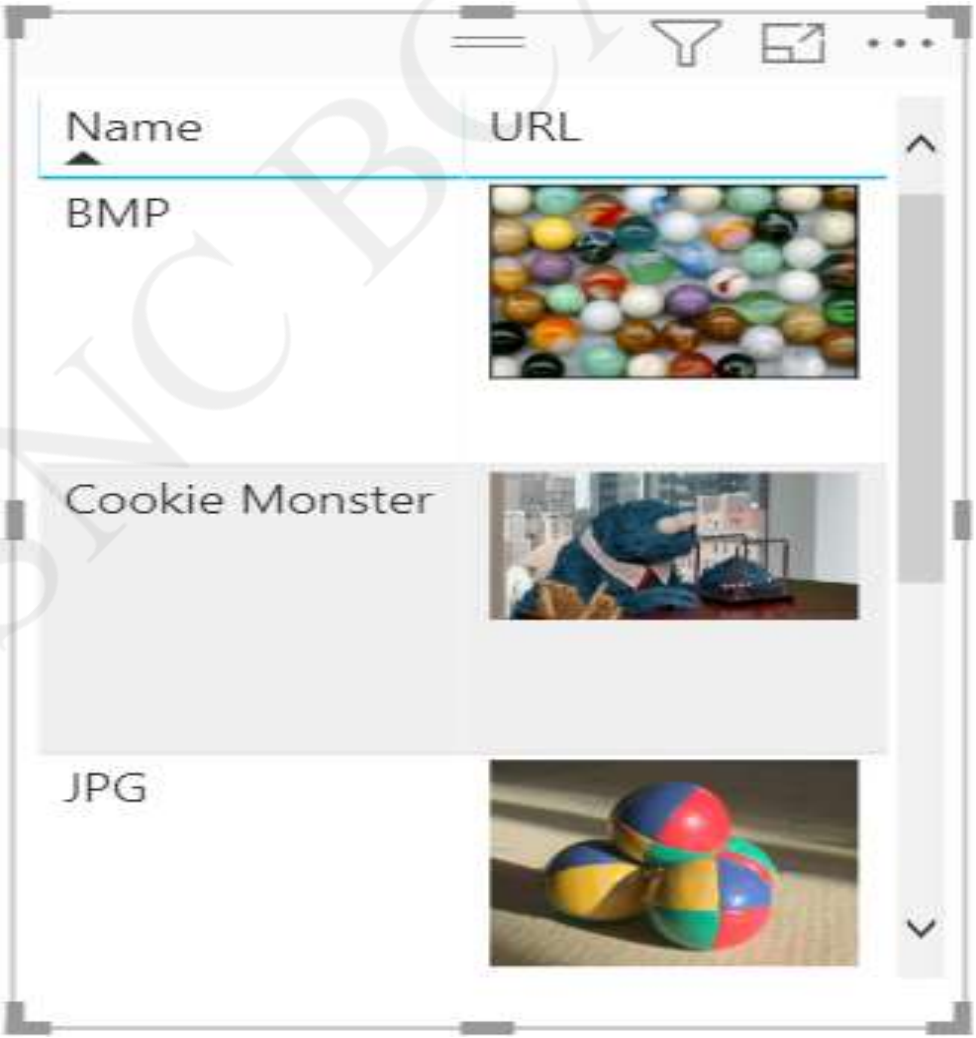
- Grouping similar documents together based on content similarity. Visualization tools can display clusters and relationships between different groups.

KLESNC BCA



Card Visuals

Image and Web URL Fields:



KLESNC BCA

Challenges of Text Visualization

1. High Dimensionality :

- Where possible use text to represent text… … which terms are the most descriptive?

2. Context & Semantics

- Provide relevant context to aid understanding. x Show (or provide access to) the source text.

3. Modeling Abstraction

- Determine your analysis task.
- Understand abstraction of your language models.
- Match analysis task with appropriate tools and models.

KLE SIC BCA

Power Query & M Language

KLESNC BCA

Power Query & M Language:

- Power Query is a data connection technology that enables you to discover, connect, combine, and refine data across a wide variety of sources.
- It is integrated into various Microsoft products, including Excel and Power BI.
- Power Query uses a functional language known as "M" to perform data transformations and manipulations.
- M is informal name of Power Query Formula Language. The formal name is so long that no one uses that, everyone call it M! M stands for Data Mashup, some say stands for Data Modeling.
- M is a functional language, which means it is designed to work with expressions and functions.
- Power Query allows you to connect to a wide range of data sources, including databases, spreadsheets, web services, and more.
- You can transform and clean your data using a graphical user interface in Power Query. The transformations are recorded and executed using the M language.

KLE SSC BCA

- After transforming the data, you can load it into the desired destination, such as a worksheet in Excel or a data model in Power BI.
- Power Query tries to push back as much data transformation work as possible to the data source. This process is known as query folding, and it helps optimize performance.
- The Power Query Editor is where you design and modify your queries. It provides a user-friendly interface for performing transformations.
- The Advanced Editor allows you to view and edit the M code directly, providing more control and flexibility for advanced users.

KLESIC BCA

How does it works

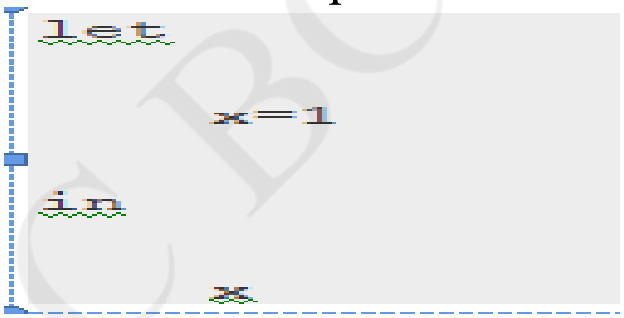


- **Extract** – Load the data from different sources
- **Transform** – Manipulate the data (replace values, pivot, add columns, etc.)
- **Load** – Export the data
- **M Engine** – Makes many decisions for us

KLE SNC BCA

Syntax of M :

- Syntax of this language is simple. it always has two blocks of programming: LET expression block, and IN expression block. Here is the most simple M syntax;



- **let** and **in** are reserved words. before going even one step further, the first and foremost thing you need to know;
- *M (Power Query Formula Language) is Case Sensitive. There is a difference between x and X.*
- what are these two programming blocks:
- **let:** definition of all variables
- **in:** output! Yes, in actually means out! just named as in. everything you put in this block will be the output of your query.

KLESIC BCA

Thank YOU

All the Best

KLESNC BCA